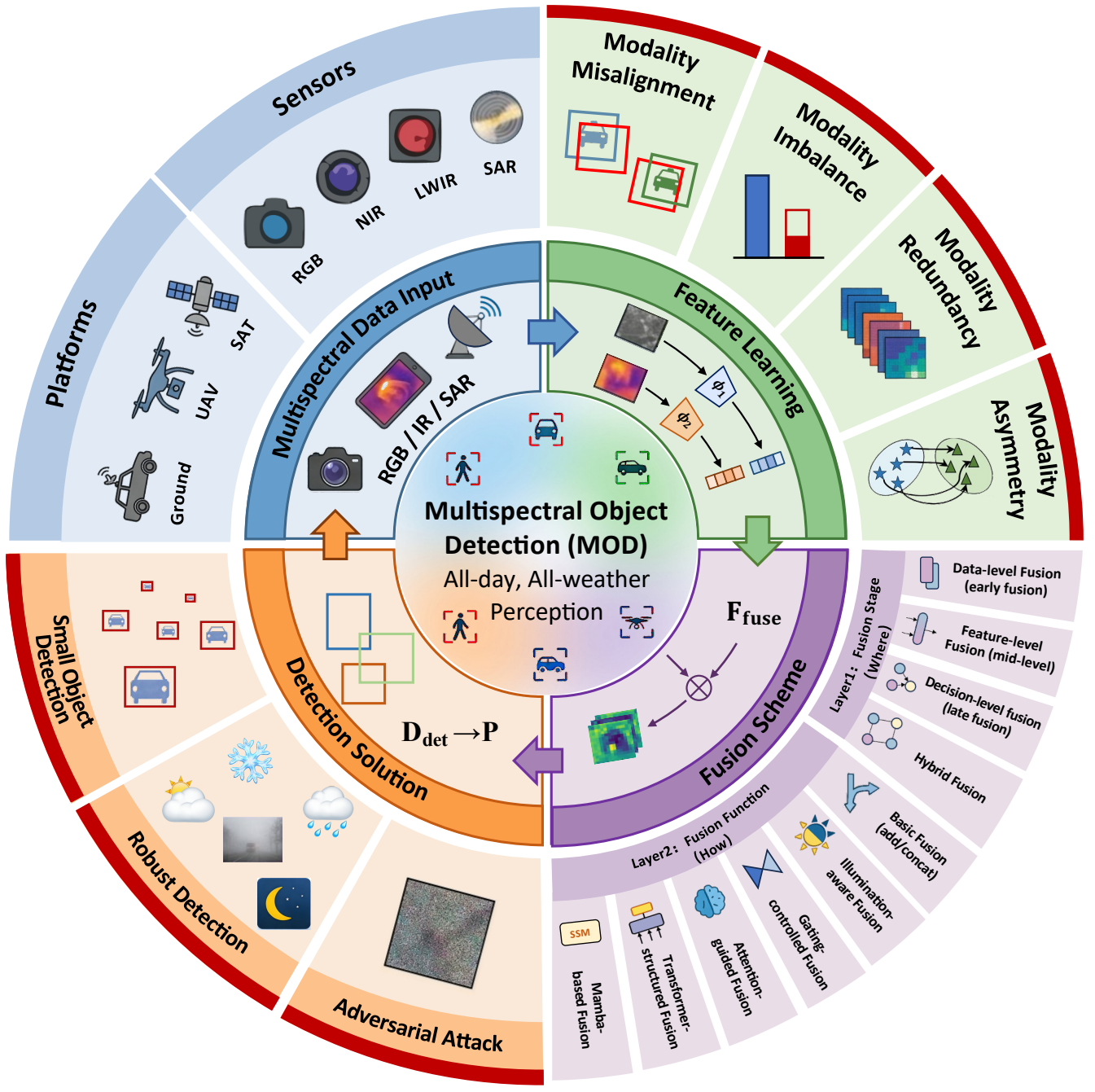
Sensors
RGB
NIR
LWIR
SAR
Platforms
SAT
UAV
Ground
Modality Misalignment
Modality Imbalance
Modality Redundancy
Modality Asymmetry
Multispectral Data Input
RGB / IR / SAR
Feature Learning
Multispectral Object Detection (MOD)
All-day, All-weather Perception
Small Object Detection
Robust Detection
Adversarial Attack
Detection Solution
$D_{det} \rightarrow P$
F_{fuse}
Fusion Scheme
Layer1: Fusion Stage (Where)
Data-level Fusion (early fusion)
Feature-level Fusion (mid-level)
Decision-level fusion (late fusion)
Hybrid Fusion
Layer2: Fusion Function (How)
Basic Fusion (add/concat)
Illumination-aware Fusion
Gating-controlled Fusion
Attention-guided Fusion
Transformer-structured Fusion
Mamba-based Fusion
SSM